



3.0 Route

3.1 Performance

Performance calculations are made before every flight and based on the information available in the aircraft POH/AFM.

3.1.1 Runway Length Calculation

To calculate the required runway length, SP's on solo flight shall factor the figures found in the POH/AOM as follows:

- (a) Take-off distance as per POH x 1.33 at least equal to or less than the take-off distance available.
- (b) Landing distance as per POH x 1.43 at least equal to or less than the landing distance available.

The SP shall take into account:

- (a) The mass of the airplane at the commencement of the take-off run at the estimated time of landing;
- (b) The pressure altitude at the aerodrome;
- (c) The ambient temperature at the aerodrome;
- (d) The runway surface condition and the type of runway surface (see table below);
- (e) The runway slope in the direction of take-off/landing (see table below).

When the POH/AOM does not include correction factors for grass runways, paved wet runways or slope the figures as determined in accordance with a, b and c above shall be multiplied as follows:

(1) TAKE-OFF

Grass (on firm soil)	DRY	1,20
up to 13cm long	WET	1,25
Paved	WET	1,15
Slope	per 1% up	5%

(2) LANDING

Grass (on firm soil)	DRY	1,20
up to 13cm long	WET	1,38
Paved	WET	1,15
Slope	per 1% down	5%

Note 1: The soil is firm when there are wheel impressions but no rutting

Note 2: slope in excess of 2% not allowed.

3.2 Flight Planning

Flight planning consists of the following items:

- Fuel calculation
- Determine the Minimum Safe Altitude (MSA)
- Navigation plan completion
- Flight Plan filling (if applicable)
- Mass and Balance and T/O and landing distance calculations
- NOTAM's
- Weather considerations
- Map preparation if required